5.1.1 In Cem 5.1.1 $\alpha \in GL_2^+(\mathbb{Q}) \Rightarrow \alpha^{-1}\Gamma\alpha \cap SL_2(\mathbb{Z})$ be congruence subgroup

need $\alpha^{-1}\Gamma\alpha$ lie in $SL_2(\mathbb{Z})$?

Sol: No. Consider $\alpha = \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix} \in GL_2^+(\mathbb{Q})$. $\alpha^{-1} = \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{pmatrix}$, $\gamma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in SL_2(\mathbb{Z})$

$$\alpha^{-1}\gamma\alpha = \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & \frac{1}{2} \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & \frac{5}{4} \\ 1 & \frac{1}{2} \end{pmatrix} \notin SL_2(\mathbb{Z}).$$

5.1.2 Show for congruence subgroups $G_1, G_2 \leq SL_2(\mathbb{Z})$. $[G_1 : G_1 \cap G_2]$, $[G_2 : G_1 \cap G_2] < \infty$.

Pf: Let $\Gamma(N_1) \leq G_1 \leq SL_2(\mathbb{Z})$. $\Gamma(N_2) \leq G_2 \leq SL_2(\mathbb{Z})$.

then $\Gamma(N_3) \leq G_1 \cap G_2 \leq SL_2(\mathbb{Z})$, $N_3 = \operatorname{lcm}(N_1, N_2)$.

Thus $[G_1 : G_1 \cap G_2] < [SL_2(\mathbb{Z}) : \Gamma(N_3)] = N_3^3 \prod_{p \mid N_3} (1 - \frac{1}{p^2}) < \infty$.

5.1.3 Show the weight-k double coset operator $[\Gamma_1 \alpha \Gamma_2]_k$ is independent of how the orbit representatives β_j are chosen.

Pf: If $\Gamma_1 \alpha \Gamma_2 = \bigcup_j \Gamma_1 \beta_j = \bigcup_j \Gamma_1 \beta_j'$

then $\exists\, \gamma_1 \in \Gamma_1$ $\gamma_1 \beta_j = \beta_j'$.

Thus $f[\beta_j']_k = f[\gamma_1 \beta_j]_k = f[\gamma_1]_k[\beta_j]_k$

Since $f \in \mathcal{M}_k(\Gamma_1)$, $f[\gamma_1]_k = f$.

Thus $f[\Gamma_1 \alpha \Gamma_2]_k = \sum_j f[\beta_j]_k = \sum_j f[\beta_j']_k$.

J.1.4. Show if $g_1, \dots, g_d : \mathcal{H} \to \mathbb{C}$ are holomorphic at ∞ then so is their sum $g_1 + \dots + g_d$.

Pf: If $g_1, \dots, g_d$ are holomorphic at ∞ then they have Fourier expansions

$$g_j(\tau) = \sum_{n \geq 0} a_n(g_j) e^{2\pi i n \tau / h_j} \quad \forall j = 1, \dots, d.$$

Let $h = \operatorname{lcm}(h_1, \dots, h_d)$, so each of the Fourier expansion would be

$$g_j(\tau) = \cancel{\sum_{n \geq 0} a_n'(g_j) e^{2\pi i n \tau / h}} \qquad a_m'(g_j) = \begin{cases} a_n(g_j) & m = n \frac{h}{h_j} \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{n \geq 0} a_n(g_j) e^{2\pi i (n h/h_j) \tau / h}$$

$$= \sum_{m \geq 0} a_m'(g_j) e^{2\pi i m \tau / h}$$

Thus $(g_1 + \dots + g_d)(\tau) = \sum_{m \geq 0} \left(a_m'(g_1) + \dots + a_m'(g_d) \right) e^{2\pi i m \tau / h}$.

is its Fourier expansion. Thus is holomorphic at ∞.

J.1.5 Check $\alpha : X_3 \to X_3'$ $\Gamma_3 \tau \mapsto \Gamma_3' \alpha(\tau)$ is well-defined.

Pf: Recall $\Gamma_3' = \alpha \Gamma_3 \alpha^{-1}$. If $\Gamma_3 \tau = \Gamma_3 \tau'$

then $\exists \gamma_3 \in \Gamma_3$ $\gamma_3(\tau) = \tau'$. Thus $\gamma_3 \alpha^{-1} \alpha(\tau) = \tau'$.

Hence $\alpha \gamma_3 \alpha^{-1} \alpha(\tau) = \alpha(\tau')$. i.e. $\Gamma_3' \alpha(\tau) = \Gamma_3' \alpha(\tau')$.

J.1.6 Show $\Gamma_1 \supset \Gamma_2$, $\alpha^{-1} \Gamma_1 \alpha = \Gamma_2$, and $\Gamma_1 \subset \Gamma_2$ the divisor group interpretation of $[\Gamma_1 \alpha \Gamma_2]_k$ gives a surjection, isomorphism, and injection.

Pf: $\Gamma_1 \supset \Gamma_2 = \Gamma_1 \alpha \Gamma_2 \overset{\alpha = I}{=} \Gamma_1$.

$$\begin{tikzcd} \Gamma_3 \arrow[r, "\simeq"] \arrow[d] & \Gamma_3' \arrow[d] \\ \Gamma_2 \arrow[r] & \Gamma_1 \end{tikzcd} \Rightarrow \Gamma_2 \tau \to \Gamma_1$$